

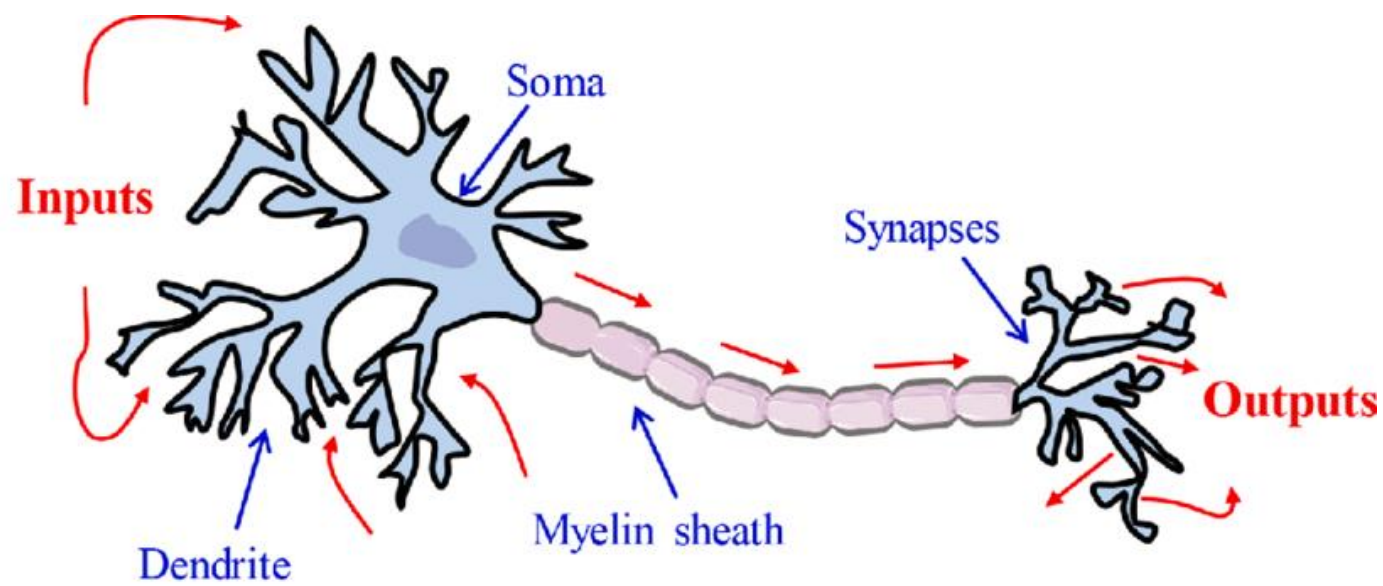
Artificial Neural Network

Natural Language Processing

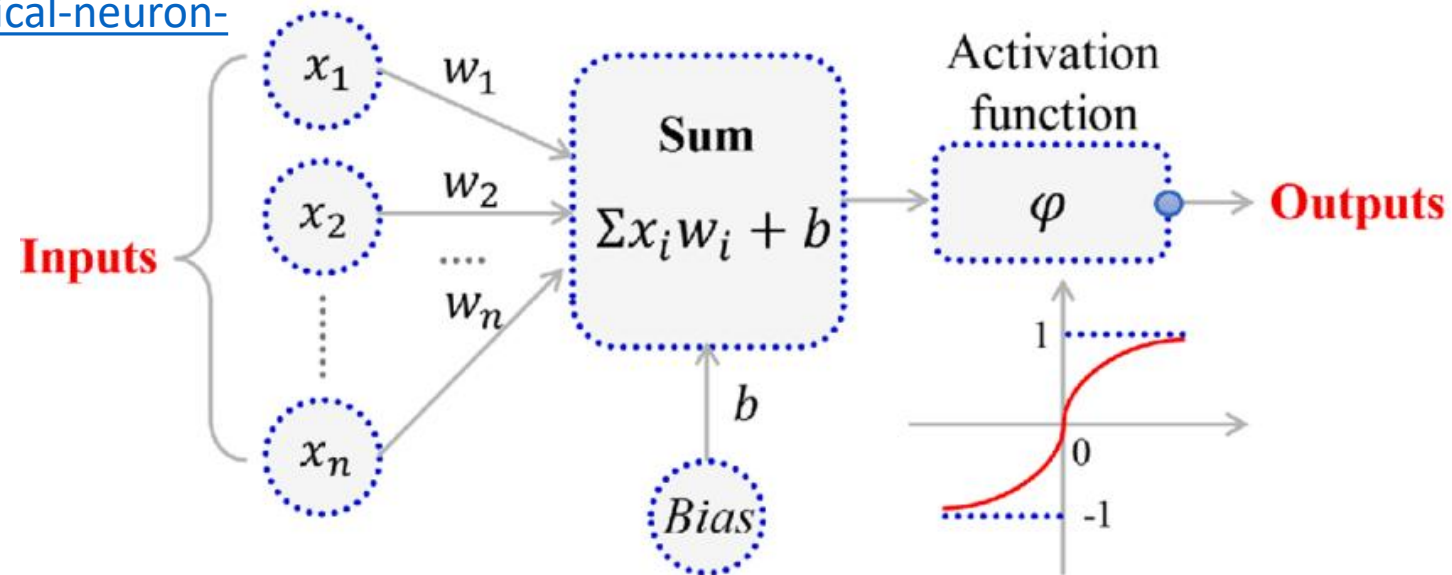
Ayesha Enayet

Definition

- A neural network is a method in artificial intelligence that teaches computers to process data in a way that is inspired by the human brain. It is a type of [machine learning](#) process that uses interconnected nodes or neurons in a layered structure that resembles the human brain. It creates an adaptive system that computers use to learn from their mistakes and improve continuously.
- <https://aws.amazon.com/what-is/neural-network/>



(a) Biological neuron



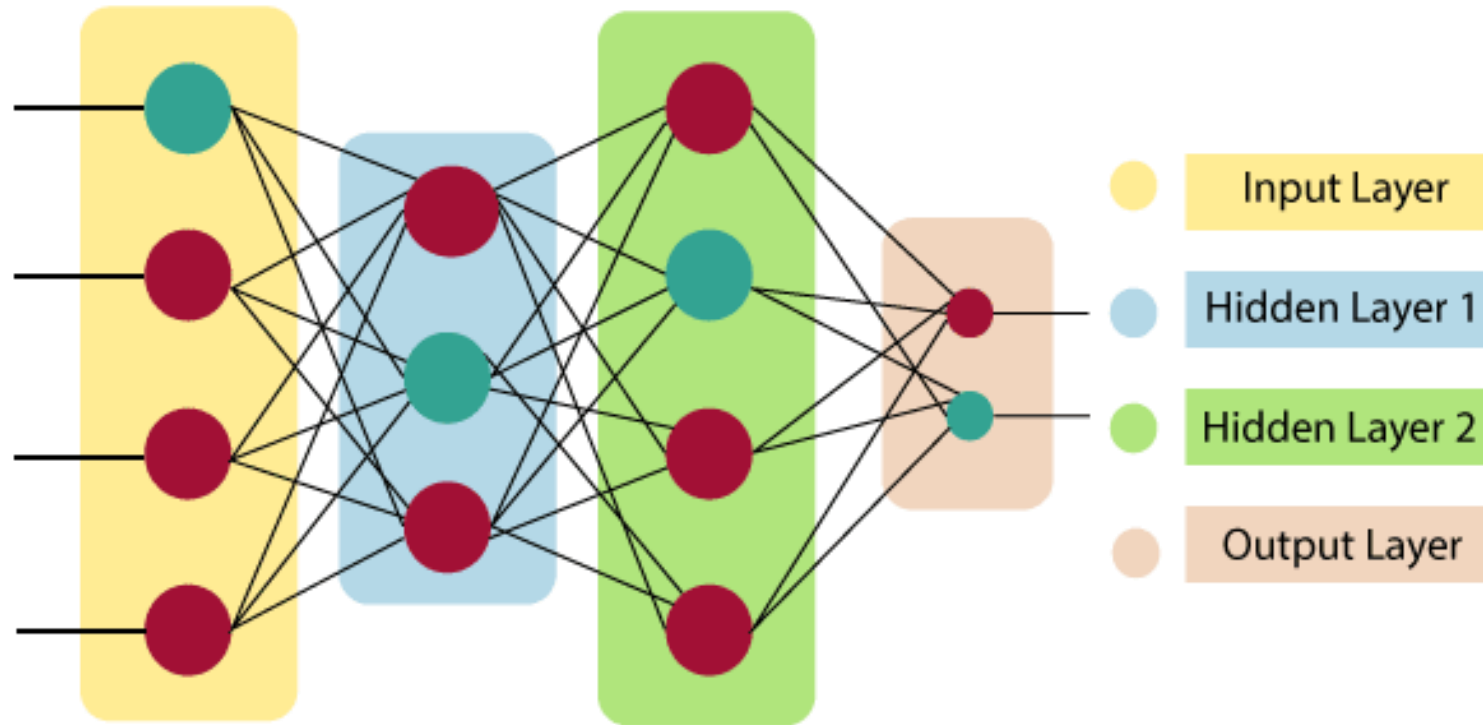
(b) Artificial neuron

Taken from

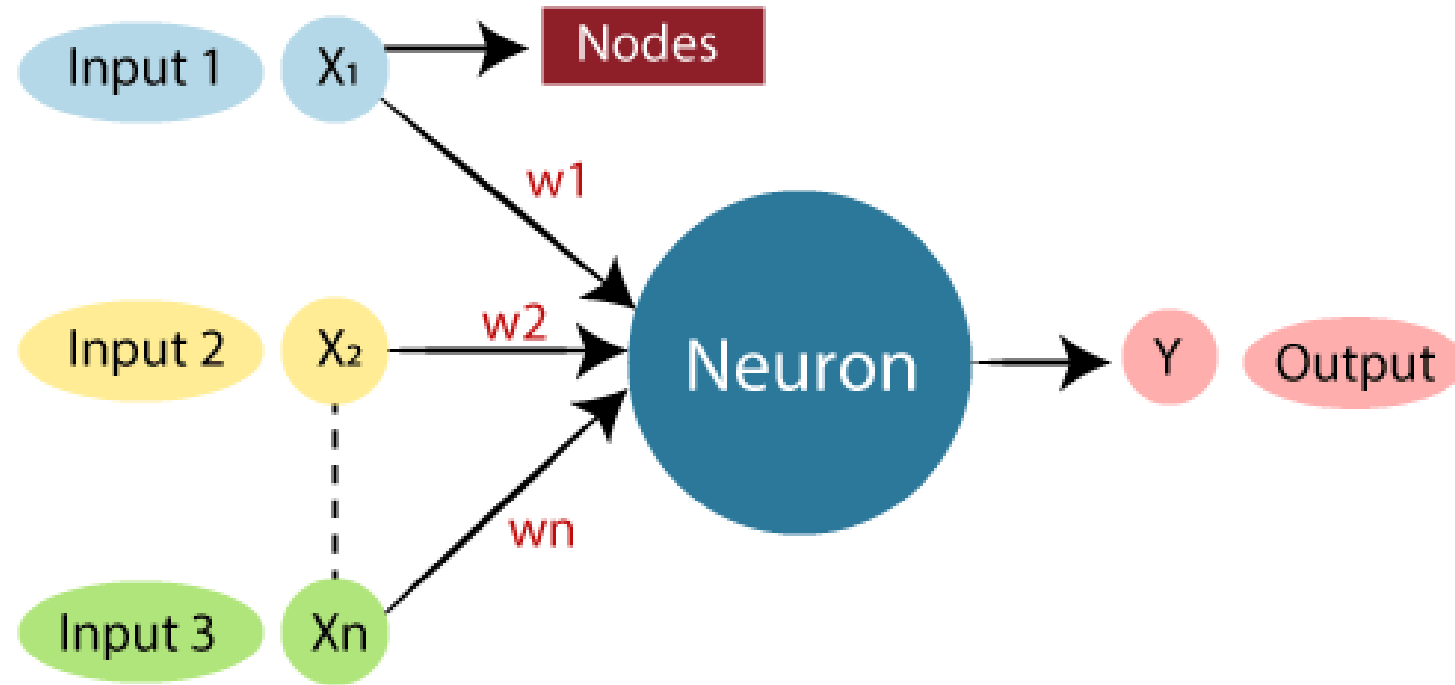
https://www.researchgate.net/figure/Comparison-between-biological-neuron-and-artificial-neuron-40_fig4_351372032

Architecture

- Figure is taken from <https://www.javatpoint.com/artificial-neural-network>



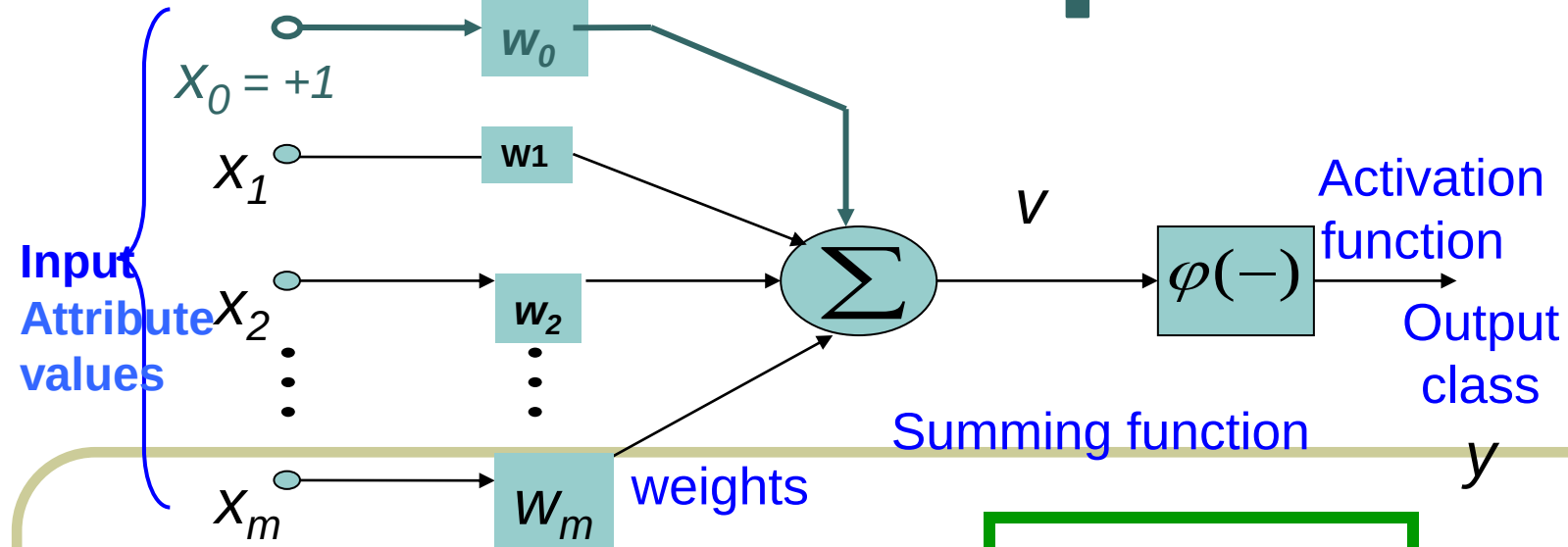
One neuron as a network



Working of one neuron

Credit : Lecture slides from **State University of New York at Stony Brook**

Bias as extra input



$$v = \sum_{j=0}^m w_j x_j$$
$$w_0 = b$$

Neuron with Activation

- The neuron is the basic information processing unit of a NN. It consists of:

- 1 A set of **links**, describing the neuron inputs, with **weights** $W_1, W_2, \dots, W_m$
2. An **adder** function (linear combiner) for computing the weighted sum of the inputs (real numbers):

$$u = \sum_{j=1}^m w_j x_j$$

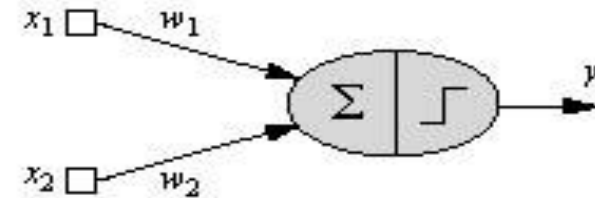


Fig1: an artificial neuron

- 3 **Activation function** : for limiting the amplitude of the neuron output.

$$y = \varphi(u + b)$$

Example

x1	x2	w1	w2	b	X (activation)
0.5	0.3	0.5	0.5	0.1	0.622

Weighted Sum = $0.5 * 0.5 + 0.3 * 0.5 + 0.1 = 0.5$

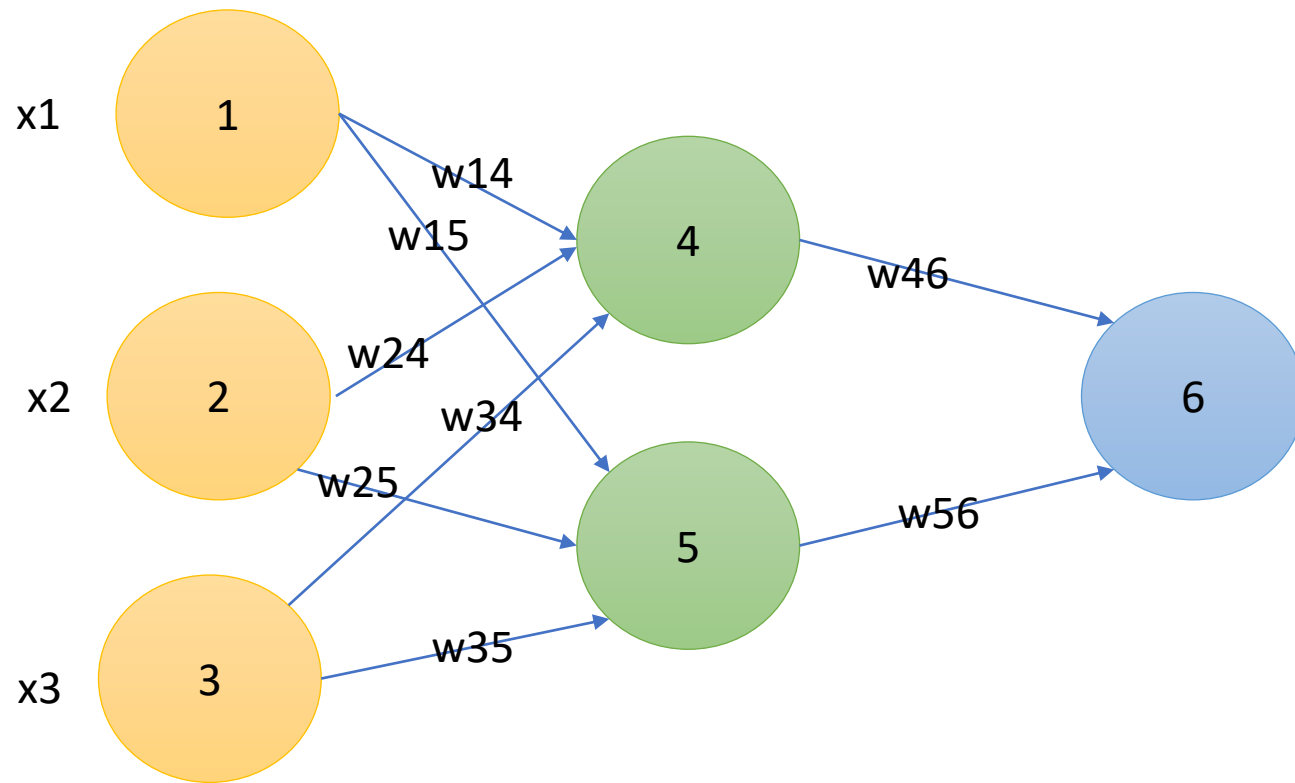
Activation function = $\frac{1}{1 + e^{-x}}$ (sigmoid function)

$Y = \Phi(X)$

$\Phi(X) = 1$ {where $X < 0.5$ }

$\Phi(X) = 0$ {where $X \geq 0.5$ }

Multi-layer neural network



Random Initialization of weights between -1 to 1
 Random Initialization of bias between -1 and 1

Activation function= $\frac{1}{1+e^{-x}}$
 (sigmoid function)

Input			H1						O		ϕ
x1	x2	x3	w14	w15	w24	W25	w34	w35	w46	w56	
0.5	0.2	0.6	-0.2	0.3	0.4	0.4	-0.5	0.4	0.6	0.2	

b4	b5	b6
-0.5	0.4	0.2

Neuron	Output O
4 $0.5 \cdot -0.2 + 0.2 \cdot 0.4 + 0.6 \cdot -0.5 + (-0.5) = -0.1 + 0.08 - 0.3 - 0.5 = -0.82$	0.305
5 $0.5 \cdot 0.3 + 0.2 \cdot 0.4 + 0.6 \cdot 0.4 + (0.4) = .15 + 0.08 + 0.24 + 0.4 = 0.87$	0.705
6 $0.305 \cdot 0.6 + 0.705 \cdot 0.2 + 0.2 = 0.524$	0.628

Input			H1						O		ϕ
x1	x2	x3	w14	w15	w24	W25	w34	w35	w46	w56	
0.5	0.2	0.6	-0.2	0.3	0.4	0.4	-0.5	0.4	0.6	0.2	

b4	b5	b6
-0.5	0.4	0.2